Design Report on FSMB RECRUITMENT ASSIGNMENT - Nazib Abrar

This project is a mains voltage protector built around an Arduino Nano (ATmega328P). It measures AC voltage and load current and keeps a relay output on D2 high while both readings stay inside their bands; outside a band, D2 drops and the appliance disconnects. I did not implement the on-device configuration path (LCD and buttons) in this phase; settings change over UART while the device runs.

Measurement and trip logic

The voltage signal arrives on A0, scaled down and biased to mid-rail; current comes from an ACS712-05B on A1 (185 mV per amp, 2.5 V with no load). One ADC count is 4.887 mV, and both AC signals swing around 512 counts.

I measure voltage as true RMS: a slow digital low-pass filter tracks the DC bias, offset $\leftarrow$ offset$+(v_{\text{raw}}-\text{offset})/1024$, and every 200 ms window (ten cycles at 50 Hz) collects the squared, de-biased samples, which turn into volts through the calibration constant:

$$V_{\text{rms}} = V_{\text{CAL}} \cdot \frac{V_{\text{ref}}/1000}{1024} \cdot \sqrt{\frac{\sum_{n=1}^{N} v_{\text{filt},n}^2}{N}}, \qquad V_{\text{CAL}} = 187$$

Current comes from the window peak: the largest raw count seen becomes sensor millivolts, the 2542 mV zero-current offset comes off, and the rest is divided by the sensitivity and by $\sqrt{2}$:

$$I_{\text{peak}} = \frac{\frac{\text{peak}}{1024} \cdot 5000 - 2542}{185}, \qquad I_{\text{rms}} = \frac{I_{\text{peak}}}{\sqrt{2}}$$

The peak shortcut is cheap on an AVR but trusts the real peak to appear among the window's ~450 samples, and the offset stays a fixed constant.

Each channel has a base and a tolerance setting; the trip band and fast-gate threshold in ADC counts follow:

$$V_{\text{upper}} = V_{\text{base}}\left(1 + \tfrac{t}{100}\right), \quad V_{\text{lower}} = V_{\text{base}}\left(1 - \tfrac{t}{100}\right), \quad I_{\text{upper}} = I_{\text{base}}\left(1 + \tfrac{t}{100}\right)$$

$$c_{\text{thr}} = 512 + \frac{V_{\text{upper}}\sqrt{2}}{V_{\text{CAL}} \cdot (V_{\text{ref}}/1000/1024)}, \qquad c_{\text{thr}} = \frac{2542}{4.887} + \frac{I_{\text{upper}}\sqrt{2} \cdot 185}{4.887}$$

The fast gate runs on every loop iteration: it compares each instantaneous sample against the threshold counts and counts consecutive hits; ten in a row, about 4.4 ms, trip the device. This gate watches only the upper edge, since the sine dips below mid-rail every negative half-cycle. The slow gate runs once per window and trips when $V < V_{\text{lower}}$, $V > V_{\text{upper}}$ or $I > I_{\text{upper}}$; it stays unarmed through the first window so the unsettled boot readings cannot trip the relay. I gave current no lower bound on purpose: an unplugged load is not a fault.

Both gates write the same latched boolean that alone drives D2; it comes back high only after 2 s of continuously normal readings. Settings changes re-derive all thresholds as integers, so the hot loop never touches floating point.

UART protocol

Configuration happens over UART at 9600 8N1, one ASCII command per line ending in `\n`.

Command	Effect	Reply
`SVB <volts>` / `SVT <pct>`	set voltage base / tolerance	`OK` with the applied value
`SCB <amps>` / `SCT <pct>`	set current base / tolerance	`OK` with the applied value
`GVB` / `GVT`	read voltage base / tolerance	value
`GCB` / `GCT`	read current base / tolerance	value

Values outside $0 <$ base ≤ 500 V (or 100 A) and $0 <$ tolerance ≤ 50 % get `ERR BAD_VALUE` back, and unknown commands get `ERR UNKNOWN_CMD`. The parser is non-blocking, so typing never slows the protection logic.